

Retail Sales Performance Analytics

A Data Analytics Case Study

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Tools Used: Python (Pandas, NumPy, Matplotlib, Seaborn), Power BI Desktop, Git/GitHub

Analysis Period: June 2024 – September 2024

Dataset Size: 60 cleaned transactions (61 raw)

1. Introduction

This case study analyzes four months of transactional sales data (June–September 2024) for a retail business selling products across five categories: Storage, Furniture, Electronics, Wearables, and Accessories. The business observed a noticeable slowdown in monthly revenue over this period and required a data-driven investigation into what is driving performance — by time, region, product category, and payment channel — to support Q4 planning decisions.

This report documents the complete analytics workflow: from raw data acquisition through cleaning, exploratory analysis, dashboard construction, and final business recommendations, using the **Ask → Prepare → Process → Analyze → Share → Act** framework.

2. Ask

2.1 Business Task

Analyze 60 retail sales transactions from June–September 2024 to identify performance trends over time, regional sales strengths, product category profitability, and payment channel behavior — in order to explain the observed sales decline and recommend concrete Q4 actions.

2.2 Key Business Questions

- How has revenue trended month-over-month across the analysis period?
- Which regions and product categories generate the most (and least) revenue?
- Are there statistical outliers in transaction value that need investigation?
- Which payment methods underperform, and does that matter?
- What is driving revenue more — order price or order size?

2.3 Stakeholders

Stakeholder	Role	Interest in Analysis
Sales Director	Primary decision-maker	Wants clear diagnosis of the revenue decline and actionable next steps

Stakeholder	Role	Interest in Analysis
Executive Team	Final approver of strategy	Needs high-level KPIs (Total Sales, Profit, Margin) to assess business health
Regional Managers	Operational stakeholders	Need region-specific performance data to prioritize inventory and marketing
Data Analytics Intern (Author)	Analyst	Responsible for cleaning, analyzing, and presenting findings

3. Prepare

3.1 Data Sources

File	Description	Rows	Columns
<code>sales_fact.csv</code>	Raw transactional sales records	61	9
<code>products.csv</code>	Product dimension table (category, cost)	10	4

3.2 ROCCC Analysis

Criterion	Assessment
Reliable	Internally sourced transactional data; low bias, but very small sample
Original	Primary transaction-level data, not aggregated or third-party
Comprehensive	Covers sale ID, date, product, region, payment, quantity, and price — sufficient for the stated business questions
Current	Covers June–September 2024, directly relevant to the recent slump under investigation
Cited	Internally generated sales records; source and structure are documented in this report

3.3 Limitations

- **Small sample size:** Only 60 transactions across 4 months limits statistical confidence, particularly for category- and region-level breakdowns.
 - **No customer-level data:** Cannot distinguish new vs. repeat customers or segment by demographics.
 - **Short time window:** Four months is insufficient to separate genuine trend from seasonality.
 - **No external context:** Marketing spend, competitor activity, and stock-availability data are not included, limiting root-cause analysis of the sales decline.
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4. Process

4.1 Tools Selected

Tool	Purpose
Python (Pandas, NumPy)	Data cleaning, transformation, statistical analysis
Matplotlib / Seaborn	Exploratory visualizations
Power BI Desktop	Interactive dashboard, DAX measures, business-facing reporting
Git / GitHub	Version control and portfolio documentation

4.2 Data Cleaning Pipeline (Cleaning Filters Applied)

#	Step	Detail
1	Remove unnecessary column	Dropped <code>notes</code> — placeholder field with no analytical value
2	Fix data types	Converted <code>order_date</code> to datetime
3	Standardize text	<code>region</code> column standardized via <code>.str.strip().str.title()</code> (fixed "north", "SOUTH" inconsistencies)
4	Impute missing values	<code>quantity</code> nulls filled with 5 (column median); <code>unit_price</code> nulls filled with 1999
5	Correct invalid value	Negative <code>quantity</code> (-3) corrected to positive using <code>.abs()</code>
6	Recalculate derived field	<code>sales_amount</code> recomputed as <code>quantity * unit_price</code> after all fixes
7	Merge with dimension table	Left join with <code>products.csv</code> on <code>product_id</code>
8	Calculate cost & profit	<code>total_cost = quantity * unit_cost</code> ; <code>profit = sales_amount - total_cost</code>
9	Remove duplicates	Dropped 1 duplicate transaction row
10	Export	Final dataset exported as <code>sales_data_cleaned.csv</code>

Representative Code:

```
python
df = sales.copy()
df = df.drop(columns=["notes"])
df["order_date"] = pd.to_datetime(df["order_date"])
df["region"] = df["region"].str.strip().str.title()
df["quantity"] = df["quantity"].fillna(5)
df["unit_price"] = df["unit_price"].fillna(1999)
df.loc[df["quantity"] < 0, "quantity"] = df.loc[df["quantity"] < 0, "quantity"].abs()
df["quantity"] = df["quantity"].astype(int)
df["sales_amount"] = df["quantity"] * df["unit_price"]
df = df.drop_duplicates()

df = df.merge(products, on="product_id", how="left")
df["total_cost"] = df["quantity"] * df["unit_cost"]
df["profit"] = df["sales_amount"] - df["total_cost"]
df.to_csv("sales_data_cleaned.csv", index=False)
```

4.3 Data Quality Verification

Check	Before Cleaning	After Cleaning
Rows	61	60
Missing values (quantity/unit_price/sales_amount)	1 / 1 / 2	0 / 0 / 0
Duplicate rows	1	0
Region text variants	6 (West, North, South, East, north, SOUTH)	4 (clean)

[🖼️ INSERT PYTHON OUTPUT: BEFORE/AFTER CLEANING SUMMARY]

5. Analyze

5.1 Descriptive Statistics

Metric	Quantity	Unit Price (₹)	Sales Amount (₹)	Profit (₹)
Mean	4.48	6,039.00	25,030.52	10,103.02
Median (50%)	4.5	5,249.00	16,743.50	7,641.50
Std Dev	2.76	4,553.83	24,582.52	9,766.51
Min	1	799.00	799.00	349.00
Max	9	14,999.00	104,993.00	42,693.00

Finding: The mean sales amount (₹25,030.52) sits well above the median (₹16,743.50), indicating a right-skewed distribution driven by a small number of high-value transactions.

[🖼️ INSERT PYTHON HISTOGRAM: DISTRIBUTION OF SALES AMOUNT]

5.2 Correlation Analysis

Variable Pair	Correlation
Sales Amount ↔ Profit	0.998
Unit Price ↔ Sales Amount	0.656
Quantity ↔ Sales Amount	0.474
Quantity ↔ Unit Price	-0.165

Finding: Sales Amount and Profit move in near-perfect lockstep (0.998), confirming profit scales directly with revenue. Unit Price is a notably stronger revenue driver (0.656) than Quantity (0.474) — **pricing and product mix matter more than order size** in this dataset.

[🖼️ INSERT PYTHON CORRELATION HEATMAP HERE]

5.3 Outlier Detection

Sale ID	Product	Quantity	Sales Amount (₹)	IQR Upper Bound
5008	Gaming Chair 7	104,993		77,617.50
5011	Gaming Chair 6	89,994		77,617.50

Finding: Two Gaming Chair transactions exceed the IQR upper bound. Both were retained as legitimate bulk orders rather than removed as errors — a business validation step, not automatic deletion.

5.4 Monthly Trend

Month	Total Sales (₹)
June	643,016
July	376,741
August	297,530
September	184,544

Finding: Revenue declined every single month, a drop of approximately **71%** from June to September — the single most urgent pattern in the dataset.

[📊 INSERT PYTHON LINE CHART: MONTHLY SALES TREND]

5.5 Regional & Category Breakdown

Region	Total Sales (₹)	Category	Total Sales (₹)	Share
South	414,840	Storage	509,940	34%
East	410,239	Furniture	329,978	22%
West	342,917	Electronics	254,046	17%
North	333,835	Wearables	208,981	14%
		Accessories	198,886	13%

Finding: South and East together account for over half of total revenue. Storage is the single largest category, contributing over a third of all sales.

5.6 Payment Method Performance

Payment Method	Total Sales (₹)
UPI	404,902
Debit Card	403,224
Cash on Delivery	398,254
Credit Card	295,451

Finding: Credit Card trails the other three methods by a consistent, meaningful margin (~₹100K+ gap from UPI).

6. Share

6.1 Power BI Dashboard Overview

The final deliverable is an interactive **Retail Sales Dashboard**, built on the same cleaned dataset (`sales_data_cleaned`) used in the Python analysis above, ensuring both tools produce identical top-line figures.

6.2 KPI Cards

KPI	DAX Measure	Result
Total Sales	<code>SUM(sales_data_cleaned[Sales Amount])</code>	₹1.50M
Total Profit	<code>SUM(sales_data_cleaned[Profit])</code>	₹606.18K
Total Transactions	<code>COUNTROWS(sales_data_cleaned)</code>	60
Profit Margin	<code>DIVIDE([Total Profit], [Total Sales Amount], 0)</code>	40.36%

6.3 Core Visuals

Visual	Type	Fields
Monthly Sales & Profit Trend	Line Chart	X: Month, Y: Total Sales, Total Profit
Sales by Category	Donut Chart	Legend: Category, Values: Total Sales
Sales by Region	Bar Chart	Y: Region, X: Total Sales
Sales by Payment Method	Column Chart	X: Payment Method, Y: Total Sales
Category × Region Matrix	Matrix	Rows: Category, Columns: Region, Values: Total Sales

Interactive Slicers: Month, Region, Category

[📷 INSERT POWER BI DASHBOARD SCREENSHOT HERE]

7. Act — Recommendations

Recommendation 1: Investigate and Reverse the Q3 Revenue Decline

- **Evidence:** Monthly sales fell every month from June (₹643,016) to September (₹184,544) — a ~71% decline, while profit margin (40.36%) remained healthy, indicating a volume/demand problem rather than a cost problem.
- **Action:** Cross-reference monthly transaction counts against marketing activity, stock availability, and seasonality using the dashboard's Month-over-Month measures to isolate whether the decline is driven by fewer transactions or smaller average order values.

- **Expected Reach/Timing:** Immediate (Q4 planning cycle) — this is the single highest-priority action, as every other recommendation is secondary to stabilizing top-line revenue.

Recommendation 2: Concentrate Inventory and Marketing on Storage in South/East

- **Evidence:** Storage is the top category (34% of revenue, ₹509,940) and South (₹414,840) / East (₹410,239) are the top two regions, together outperforming West and North significantly.
- **Action:** Prioritize stock availability and targeted promotions for Storage products specifically in South and East before expanding investment into lower-performing region-category combinations.
- **Expected Reach/Timing:** Next 4–6 weeks — low-risk, high-confidence action based on proven demand rather than speculative expansion.

Recommendation 3: Diagnose the Credit Card Payment Gap

- **Evidence:** Credit Card generates ₹295,451 versus ₹398K–₹405K for the other three payment methods — a consistent ~27% gap too large to dismiss as random variation.
 - **Action:** Audit whether Credit Card transactions face structural friction (processing fees, verification steps, slower confirmation) versus UPI/Debit Card; a targeted UX or incentive fix could recover meaningful revenue share.
 - **Expected Reach/Timing:** Within one quarter — requires coordination with payment processing/checkout teams, so timeline is moderate rather than immediate.
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8. Additional Data for Future Analysis

To strengthen future iterations of this analysis, the following data would be valuable:

- **Customer demographic data** (age, location, loyalty tier) to segment revenue drivers beyond region alone
 - **Marketing spend and campaign data** to directly test correlation with the observed monthly decline
 - **Inventory/stock-level records** to distinguish demand-driven decline from stock-availability constraints
 - **A longer historical time window** (12+ months) to separate genuine trend from seasonal effects
 - **Customer acquisition vs. repeat-purchase data** to understand whether the decline stems from fewer new customers or reduced repeat orders
 - **Competitor pricing or market data** for external context on the Q3 slowdown
 - **Payment processing fee structure** to directly test the Credit Card underperformance hypothesis
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9. Methodology & Reproducibility

Phase	Tool	Script / File	Output
Ask	—	This report, Section 2	Business task & stakeholder definition
Prepare	—	sales_fact.csv, products.csv	Raw data inventory & ROCCC assessment
Process	Python (Pandas)	Dataset_Selection_Data_Preparation.ipynb	sales_data_cleaned.csv (60 rows, 13 columns, 0 nulls, 0 duplicates)
Analyze	Python (Pandas, NumPy)	EDA_Visualizations.ipynb	Descriptive stats, correlation matrix, outlier detection, 8 visualizations
Share	Power BI Desktop	Retail Sales Dashboard (.pbix)	4 KPIs, 5 visuals, 3 slicers
Act	—	This report, Section 7	3 data-driven recommendations

GitHub Repository Link: [ INSERT GITHUB REPOSITORY LINK HERE]

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End of Case Study — Retail Sales Performance Analytics
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